

The 45-Bit Universe: An Informational Approach to Dark Energy

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Abstract

We propose that dark energy emerges from informational decoherence in a 5D de Sitter bulk. The 11D→5D collapse breaks $SO(10,1) \rightarrow SO(4,1)$ yielding $N=45$ channels. The fine-structure constant α fixes the per-channel fidelity $f = (1 - 12\alpha)^{1/45} = 0.9124$. This yields $\Omega_\Lambda = f^{45} = 0.685$ without free parameters, matching Planck 2018. The model derives the dark energy equation of state $w = -1 + 2/(3N) \approx -0.985$, distinguishable from Λ CDM with DESI 2025. This is Part 1 of the Informational Sheets Cosmology series establishing the foundational framework.

Keywords: dark energy, information theory, fine-structure constant, cosmology, 5D de Sitter, decoherence, $SO(10,1)$

1 Introduction

The cosmological constant problem remains unresolved. We propose an informational origin: vacuum energy is residual coherence from a 5D de Sitter bulk.

2 The 5D Informational Framework

An 11D Planck-scale manifold with $SO(10,1)$ symmetry collapses to 5D de Sitter with $SO(4,1)$. This yields $N = \dim(\mathfrak{so}(10,1)) - \dim(\mathfrak{so}(4,1)) = 55 - 10 = 45$ independent channels.

Each channel has fidelity f set by the fine-structure constant:

$$f = (1 - 12\alpha)^{1/45} = 0.912403 \tag{1}$$

using $\alpha^{-1} = 137.035999084$ from CODATA 2018.

3 Derivation of Dark Energy

The total bulk coherence is $f^N = f^{45}$. Projected to 4D, this becomes the dark energy fraction:

$$\Omega_\Lambda = f^{45} = 0.9124^{45} = 0.685 \quad (2)$$

This matches Planck 2018: $\Omega_\Lambda = 0.6847 \pm 0.0073$ with zero free parameters.

The equation of state follows from information dynamics:

$$w = -1 + \frac{2}{3N} = -1 + \frac{2}{135} = -0.9852 \quad (3)$$

4 Predictions and Falsifiability

1. $w_0 = -0.985$, $w_a = 0$: testable with DESI 2025. 2. No phantom crossing: $w(z) > -1$ always. 3. The same f predicts S8 and H0 tensions: see Parts 3 and 4.

5 Conclusion

Dark energy is explained as 5D informational decoherence. The fine-structure constant sets $\Omega_\Lambda = 0.685$ exactly. This is the foundation for resolving S8 and H0 tensions in subsequent papers.

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Code Availability: Analytical results reproducible with any computer algebra system.

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